

IRON SPRINGS AGDM, AB, Canada

WMO: 715280

Lat: 49.9005N

Lon: 112.7452W

Elev: 2930

StdP: 13.21

Time Zone: -7.00 (NAM)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-16.9	-11.5	-23.0	1.7	-16.8	-17.3	2.4	-11.5	36.8	41.8	33.9	42.1	7.7	260	0.708

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	25.9	86.6	62.7	83.2	61.5	80.1	60.8	66.8	79.8	64.8	78.2	62.7	75.9	8.8	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.0	92.6	74.0	59.7	85.1	71.7	57.4	78.5	68.8	33.1	80.0	31.4	78.0	29.8	75.8	75.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
34.0	29.4	25.5	-25.8	93.3	5.7	3.5	-29.9	95.8	-33.3	97.9	-36.5	99.8	-40.7	102.4
			-25.8	71.9	5.8	2.5	-29.9	73.6	-33.3	75.1	-36.5	76.5	-40.7	78.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	42.3	22.4	20.9	31.7	41.8	52.3	59.0	64.8	63.2	55.0	43.2	31.2	20.6	(6)
(7)		DBStd	19.31	15.52	15.38	13.34	9.06	7.83	5.55	4.90	5.73	8.18	9.35	13.41	14.97	(7)
(8)		HDD50	4342	855	815	570	266	66	3	0	1	42	244	569	911	(8)
(9)		HDD65	8418	1320	1234	1033	696	397	195	65	102	309	676	1015	1376	(9)
(10)		CDD50	1531	0	0	2	21	138	272	458	409	193	34	4	0	(10)
(11)		CDD65	131	0	0	0	0	4	13	58	46	10	0	0	0	(11)
(12)		CDH74	2519	0	0	0	16	168	264	925	835	303	8	0	0	(12)
(13)		CDH80	696	0	0	0	1	30	55	264	263	83	0	0	0	(13)
(14)	Wind	WSAvg	11.1	13.6	11.0	11.8	12.0	10.9	10.2	8.2	8.2	9.6	11.5	13.0	12.8	(14)
(15)	Precipitation	PrecAvg	14.8	0.5	0.4	0.6	1.1	2.5	3.6	1.6	1.4	1.3	0.9	0.6	0.6	(15)
(16)		PrecMax	23.0	1.0	0.8	1.4	2.9	4.4	9.7	5.2	3.9	4.6	2.9	1.3	1.3	(16)
(17)		PrecMin	7.7	0.2	0.0	0.0	0.4	0.7	0.8	0.1	0.0	0.2	0.2	0.1	0.0	(17)
(18)		PrecStd	4.1	0.2	0.2	0.4	0.6	1.0	2.1	1.2	0.9	1.0	0.6	0.3	0.3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.1	56.5	63.0	76.0	83.7	86.6	90.4	90.5	86.9	74.1	64.1	52.2	(19)
(20)			MCWB	42.3	42.4	45.7	52.4	56.0	63.8	65.5	61.5	58.9	52.9	48.3	41.6	(20)
(21)		2%	DB	48.7	50.0	58.7	68.5	79.0	80.3	86.4	86.4	82.2	68.0	56.4	47.6	(21)
(22)			MCWB	40.4	39.5	44.4	48.7	55.5	60.7	65.4	61.5	57.8	51.0	44.7	38.6	(22)
(23)		5%	DB	45.1	45.0	54.4	63.2	73.9	76.7	82.7	82.6	77.1	62.7	52.0	43.8	(23)
(24)			MCWB	37.8	36.4	42.0	46.0	53.6	59.5	64.3	60.5	56.4	48.8	42.5	36.6	(24)
(25)		10%	DB	41.5	41.1	49.8	58.2	69.3	72.8	79.3	78.7	71.5	58.1	48.0	40.5	(25)
(26)			MCWB	35.4	34.3	39.8	44.1	51.4	57.5	63.1	60.2	54.4	46.4	39.9	34.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.3	43.7	47.9	52.8	58.9	67.0	71.6	67.0	62.0	55.2	49.9	43.1	(27)
(28)			MCD B	52.0	53.4	60.2	71.5	78.5	79.4	83.6	81.2	79.4	69.2	62.0	50.5	(28)
(29)		2%	WB	40.6	39.7	45.2	49.7	56.4	63.7	68.3	64.7	59.2	52.1	45.9	39.1	(29)
(30)			MCD B	48.2	48.8	56.8	66.6	75.3	76.9	81.5	79.9	77.6	65.8	55.0	45.5	(30)
(31)		5%	WB	38.0	37.0	42.7	46.9	54.4	61.0	66.2	62.8	57.2	49.5	42.8	36.7	(31)
(32)			MCD B	44.3	44.6	53.1	61.0	71.4	73.5	79.0	78.0	74.5	60.9	51.2	42.7	(32)
(33)		10%	WB	35.7	34.5	40.1	44.5	52.6	58.9	64.3	61.0	55.1	47.0	40.1	34.1	(33)
(34)			MCD B	41.1	40.7	49.3	56.9	66.9	70.7	77.0	75.3	70.5	57.2	47.4	40.0	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	18.1	19.8	20.6	22.8	23.5	22.5	25.9	26.7	25.1	22.0	19.0	17.7	(35)
(36)			MCDBR	18.2	24.6	27.5	32.8	32.2	29.8	32.1	35.3	35.4	29.2	23.4	18.7	(36)
(37)			MCWBR	12.2	15.8	15.8	16.4	13.8	14.0	15.9	14.7	15.0	15.0	14.2	12.7	(37)
(38)		5% WB	MCD BR	17.9	23.8	26.5	30.4	28.8	27.3	29.4	29.9	33.1	27.7	22.5	19.3	(38)
(39)			MCWBR	12.7	16.0	16.1	16.0	13.0	14.2	15.8	14.2	14.5	15.0	14.5	13.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.251	0.260	0.284	0.320	0.335	0.332	0.334	0.351	0.312	0.280	0.256	0.235	(40)	
(41)		taud	2.462	2.496	2.487	2.389	2.400	2.455	2.463	2.395	2.476	2.542	2.507	2.460	(41)	
(42)		Ebn at Noon	262	288	297	294	293	294	291	280	282	276	259	254	(42)	
(43)		Edh at Noon	18	23	28	34	36	34	33	34	28	22	17	16	(43)	
(44)	All-Sky Solar Radiation	RadAvg	391	707	1112	1535	1830	2027	2221	1768	1255	779	432	305	(44)	
(45)		RadStd	32	58	89	104	146	142	113	110	108	82	36	29	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (69 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page